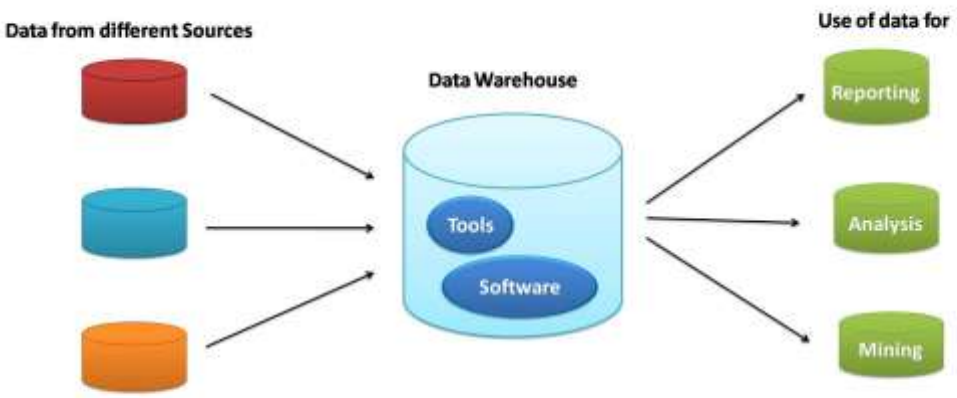
Module 1

Data Warehouse (DWH) Fundamentals with Introduction to Data Mining

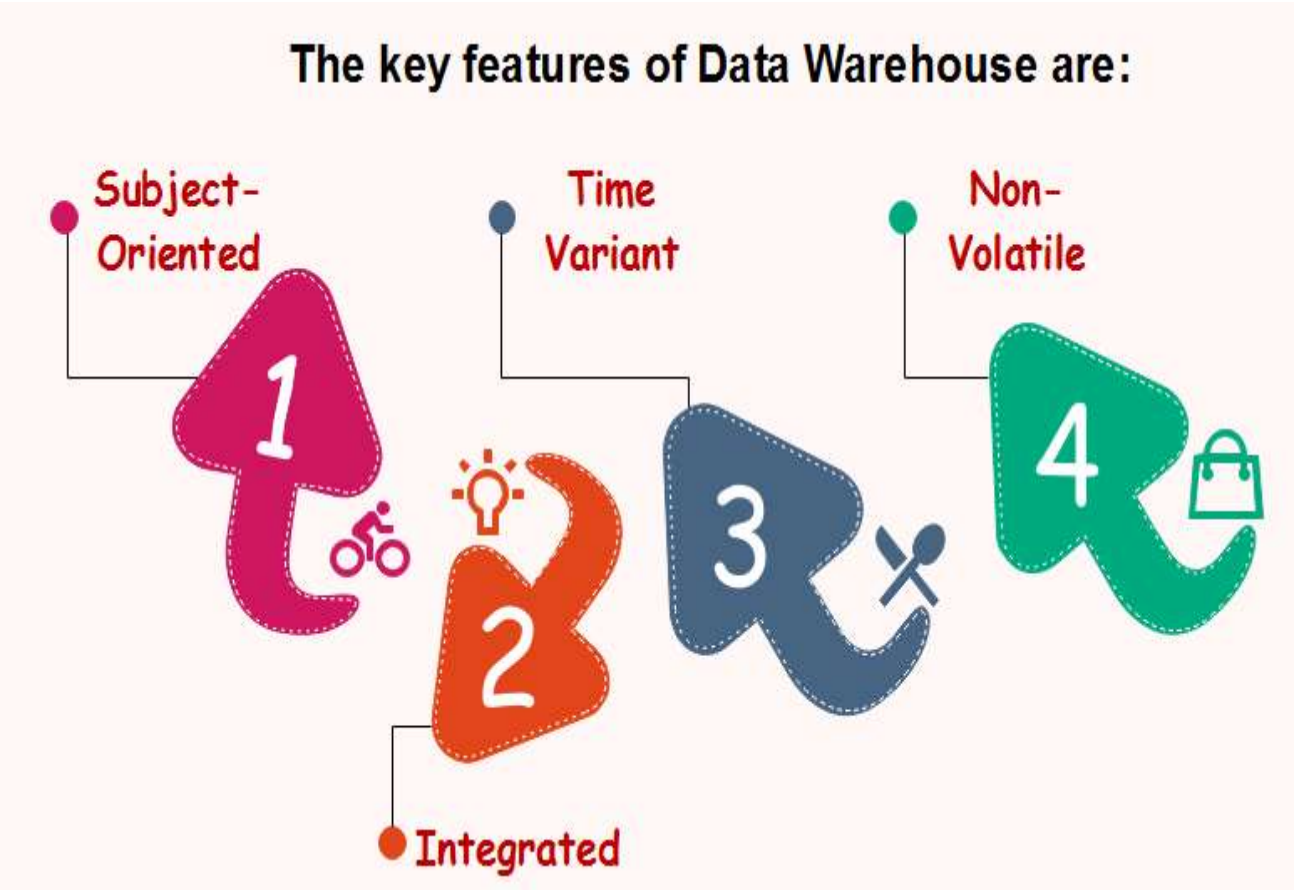
1. What is Data Warehouse?

A Data Warehouse is a collection of software tools that facilitates analysis of a large set of business data used to help an organization make decisions. A large amount of data in data warehouses comes from numerous sources such that internal applications like marketing, sales, and finance; customer-facing apps; and external partner systems, among others. It is a centralized data repository for analysts that can be queried whenever required for business benefits. A data warehouse is mainly a data management system that's designed to enable and support business intelligence (BI) activities, particularly analytics. Data warehouses are alleged to perform queries, cleaning, manipulating, transforming and analyzing the data and they also contain large amounts of historical data.



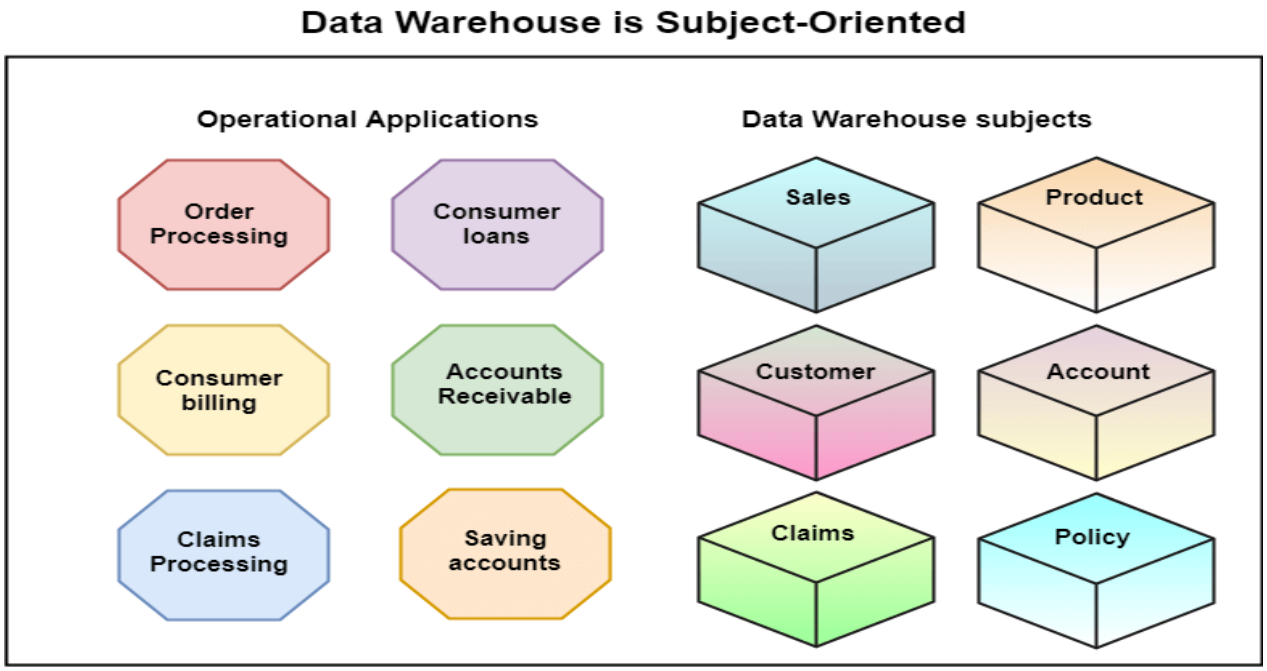
The process of creating data warehouses to store a large amount of data is named Data Warehousing. Data Warehousing helps to improve the speed and efficiency of accessing different data sets and makes it easier for company decision-makers to obtain insights that will help the business and promoting marketing tactics that set them aside from their competitors. We can say that it is a blend of technologies and components which aids the strategic use of data and information. The main goal of data warehousing is to create a hoarded wealth of historical data that can be retrieved and analyzed to supply helpful insight into the organization's operations.

2. Characteristics of Data Warehouse



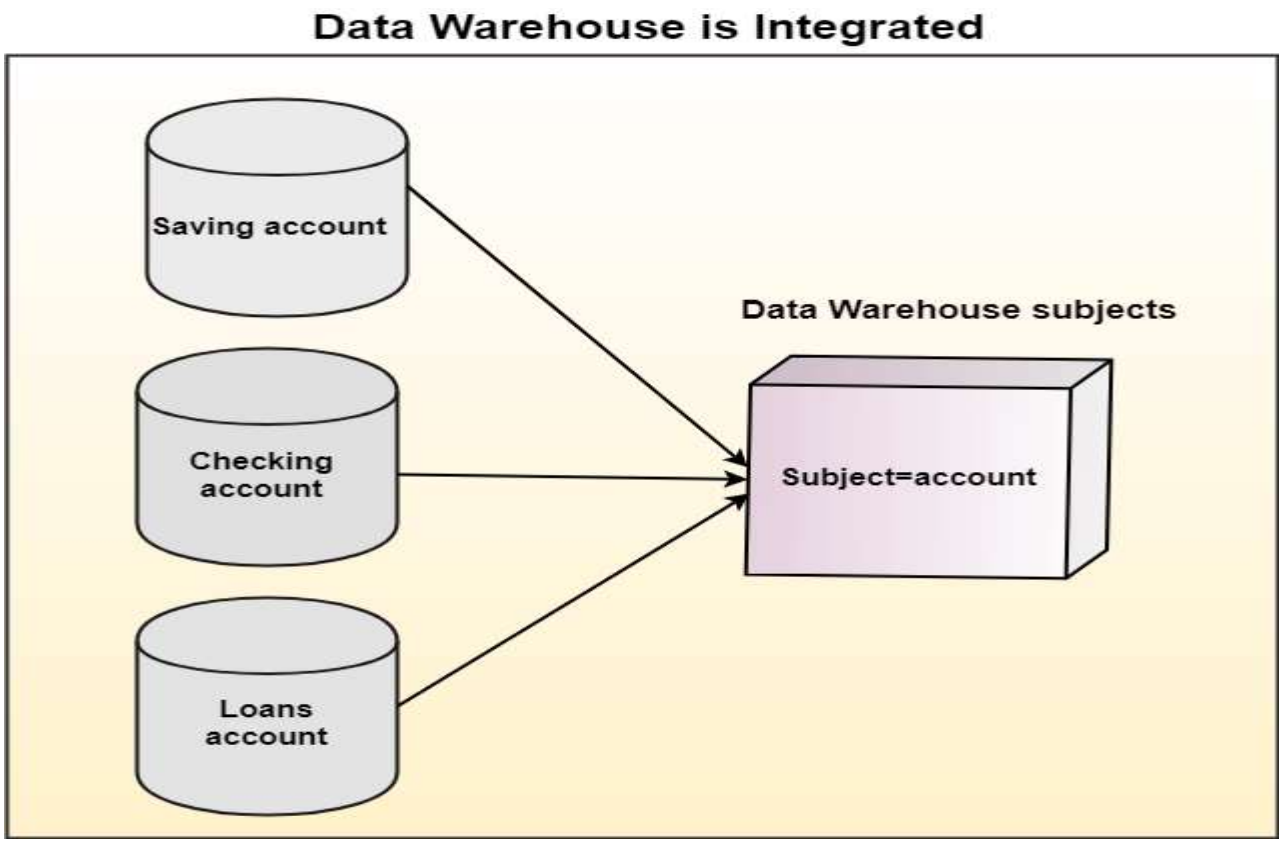
Subject-Oriented

A data warehouse target on the modeling and analysis of data for decision-makers. Therefore, data warehouses typically provide a concise and straightforward view around a particular subject, such as customer, product, or sales, instead of the global organization's ongoing operations. This is done by excluding data that are not useful concerning the subject and including all data needed by the users to understand the subject.



Integrated

A data warehouse integrates various heterogeneous data sources like RDBMS, flat files, and online transaction records. It requires performing data cleaning and integration during data warehousing to ensure consistency in naming conventions, attributes types, etc., among different data sources.

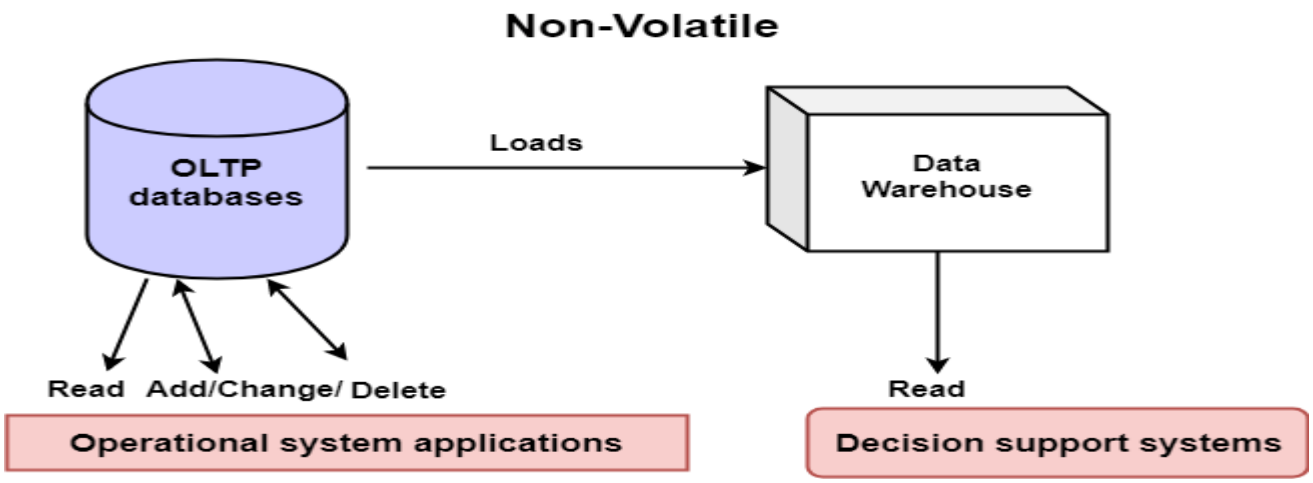


Time-Variant

Historical information is kept in a data warehouse. For example, one can retrieve files from 3 months, 6 months, 12 months, or even previous data from a data warehouse. These variations with a transactions system, where often only the most current file is kept.

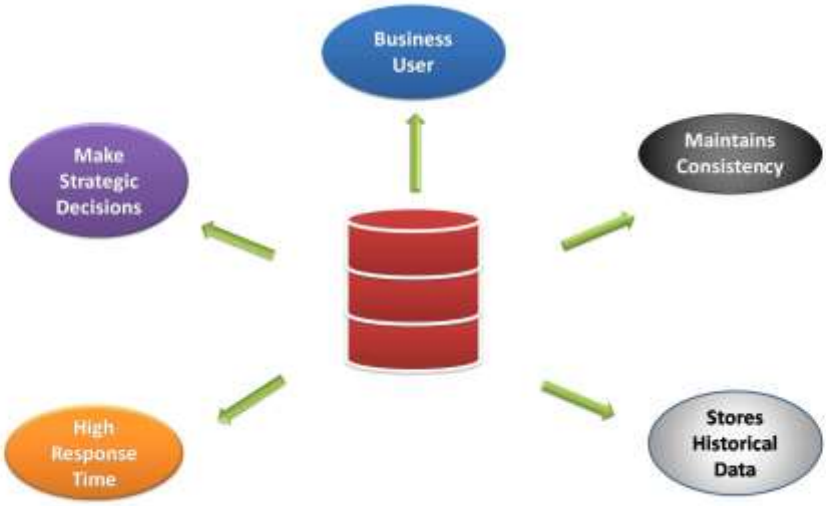
Non-Volatile

The data warehouse is a physically separate data storage, which is transformed from the source operational RDBMS. The operational updates of data do not occur in the data warehouse, i.e., update, insert, and delete operations are not performed. It usually requires only two procedures in data accessing: Initial loading of data and access to data. Therefore, the DW does not require transaction processing, recovery, and concurrency capabilities, which allows for substantial speedup of data retrieval. Non-Volatile defines that once entered into the warehouse, and data should not change.



3. Need of Data Warehousing.

Data Warehousing is a progressively essential tool for business intelligence. It allows organizations to make quality business decisions. The data warehouse benefits by improving data analytics, it also helps to gain considerable revenue and the strength to compete more strategically in the market. By efficiently providing systematic, contextual data to the business intelligence tool of an organization, the data warehouses can find out more practical business strategies.



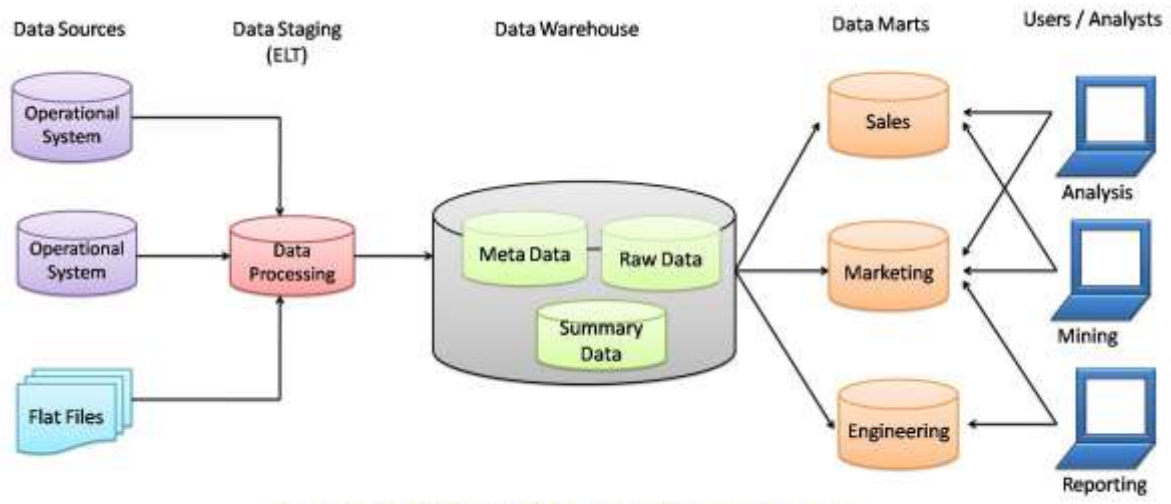
1. **Business User:** Business users or customers need a data warehouse to look at summarized data from the past. Since these people are coming from a non-technical background also, the data may be represented to them in an uncomplicated way.
2. **Maintains consistency:** Data warehouses are programmed in such a way that they can be applied in a regular format to all collected data from different sources, which makes it effortless for company decision-makers to analyze and share data insights with their colleagues around the globe. By standardizing the data, the risk of error in interpretation is also reduced and improves overall accuracy.
3. **Store historical data:** Data Warehouses are also used to store historical data that means, the time variable data from the past and this input can be used for various purposes.
4. **Make strategic decisions:** Data warehouses contribute to making better strategic decisions. Some business strategies may be depending upon the data stored within the data warehouses.
5. **High response time:** Data warehouse has got to be prepared for somewhat sudden masses and type of queries that demands a major degree of flexibility and fast latency.

4. Architecture & Components of Data Warehouse:

Data warehouse architecture defines the comprehensive architecture of data processing and presentation that will be useful for data analysis and decision making within the enterprise and organization. Each organization has different data warehouses depending upon their need, but all of them are characterized by some standard components.

Data Warehouse applications are designed to support the user's data requirements, an example of this is online analytical processing (OLAP). These include functions such as forecasting, profiling, summary reporting, and trend analysis.

The architecture of the data warehouse mainly consists of the proper arrangement of its elements, to build an efficient data warehouse with software and hardware components. The elements and components may vary based on the requirement of organizations. All of these depend on the organization's circumstances.



DATA WAREHOUSE ARCHITECTURE

1. Source Data Component:

In the Data Warehouse, the source data comes from different places. They are group into four categories:

- **External Data:** For data gathering, most of the executives and data analysts rely on information coming from external sources for a numerous amount of the information they use. They use statistical features associated with their organization that is brought out by some external sources and department.
- **Internal Data:** In every organization, the consumer keeps their "private" spreadsheets, reports, client profiles, and generally even department databases. This is often the interior information, a part that might be helpful in every data warehouse.

- **Operational System data:** Operational systems are principally meant to run the business. In each operation system, we periodically take the old data and store it in achieved files.
- **Flat files:** A flat file is nothing but a text database that stores data in a plain text format. Flat files generally are text files that have all data processing and structure markup removed. A flat file contains a table with a single record per line.

2. Data Staging:

After the data is extracted from various sources, now it's time to prepare the data files for storing in the data warehouse. The extracted data collected from various sources must be transformed and made ready in a format that is suitable to be saved in the data warehouse for querying and analysis. The data staging contains three primary functions that take place in this part:

- **Data Extraction:** This stage handles various data sources. Data analysts should employ suitable techniques for every data source.
- **Data Transformation:** As we all know, information for a knowledge warehouse comes from many alternative sources. If information extraction for a data warehouse posture huge challenges, information transformation gifts even important challenges. We tend to perform many individual tasks as a part of information transformation. First, we tend to clean the info extracted from every source of data. Standardization of information elements forms an outsized part of data transformation. Data transformation contains several kinds of combining items of information from totally different sources. Information transformation additionally contains purging supply information that's not helpful and separating outsourced records into new mixtures. Once the data transformation performs ends, we've got a set of integrated information that's clean, standardized, and summarized.
- **Data Loading:** When we complete the structure and construction of the data warehouse and go live for the first time, we do the initial loading of the data into the data warehouse storage. The initial load moves high volumes of data consuming a considerable amount of time.

3. Data Storage in Warehouse:

Data storage for data warehousing is split into multiple repositories. These data repositories contain structured data in a very highly normalized form for fast and efficient processing.

- **Metadata:** Metadata means data about data i.e. it summarizes basic details regarding data, creating findings & operating with explicit instances of data. Metadata is generated by an additional correction or automatically and can contain basic information about data.
- **Raw Data:** Raw data is a set of data and information that has not yet been processed and was delivered from a particular data entity to the data supplier and hasn't been processed nonetheless by machine or human. This data is gathered out from online sources to deliver deep insight into users' online behavior.
- **Summary Data or Data summary:** Data summary is an easy term for a brief conclusion of an enormous theory or a paragraph. This is often one thing where analysts write the code and in the end, they declare the ultimate end in the form of summarizing data. Data summary is the most essential thing in data mining and processing.

4. Data Marts:

Data marts are also the part of storage component in a data warehouse. It can store the information of a specific function of an organization that is handled by a single authority. There may be any number of data marts in a particular organization depending upon the functions. In short, data marts contain subsets of the data stored in data warehouses.